Coefficient of Friction & F = ma

Exam code: 9MA0

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Coefficient of friction & F = ma

How do I use F = ma in problems involving friction?

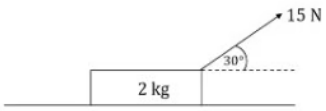
- The **coefficient of friction** combined with ***F = ma*** allows you to determine an object's motion where friction is involved in a problem
- For problems where the surface is horizontal:
- Step 1. If necessary, **resolve** any angled forces into vertical and horizontal components
- Step 2. Calculate the normal reaction force
  - Be careful – if there are vertical forces other than gravity these will affect the value of ***R***
  - with a horizontal surface ***R*** will always be directed vertically upwards
  - the magnitude of ***R*** will be such as to make the **total** vertical force on the object zero
- Step 3. Calculate $F_{MAX} = \mu R$ and find the **resultant** (total force) of all the horizontal forces on the object
  - Remember – if the resultant of the other horizontal forces is less than or equal to F_{MAX} then friction will exactly balance those forces out and the object will remain stationary
- Step 4. Use ***F = ma*** to determine the acceleration of the object

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

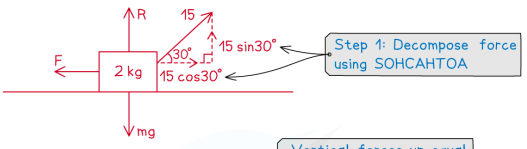
A wooden block of mass 2 kg is at rest on a rough horizontal floor, where the coefficient of friction between the block and the floor is 0.6. A constant force of 15 N is applied to the block at an angle of 30° to the horizontal, as shown in the diagram below.



Find the acceleration of the block.

Answer:

Start by drawing a diagram



Vertical forces up equal to vertical forces down

Step 2: $R + 15\sin 30^\circ = mg$

$R = mg - 15\sin 30^\circ = 2 \times 9.8 - 15 \times \frac{1}{2}$

$R = 12.1\text{ N}$

Step 3: $F_{max} = \mu R = 0.6 \times 12.1 = 7.26\text{ N}$

$15\cos 30 = \frac{15\sqrt{3}}{2} = 12.99...\text{ N} > F_{max}$

so $F = F_{max}$ and total horizontal force is

$12.99... - 7.26 = 5.73...\text{ N}$ (to the right)

Step 4: $F_{total} = ma$

$5.73... = 2a$

$a = 2.87\text{ ms}^{-2}$ (3 sf)

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Examiner Tips and Tricks

- Always draw a force diagram and label it clearly. Look out for the words **smooth** and **rough** in mechanics problems involving an object moving (or potentially moving) along a surface:
  - If the surface is described as **smooth** then you can ignore friction in the problem (ie ***μ= 0***)
  - If the surface is described as **rough** than you need to include the force of friction in solving the problem
- Be aware of whether the question is on a horizontal surface or an **inclined plane.**

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